

# Scalability & Future Plan

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|---------------|------------------------------------------------------|
| Date          | 12 July 2026                                         |
| Team ID       | 6a2fbf036b1ed0bd8fa789aa                             |
| Project Name  | AI Powered Debt Relief & Financial Recovery Platform |
| Maximum Marks | 1 Mark                                               |

## Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

## Current System Limitations:

| S.No | Limitation                                                        | Impact                                                                            | Priority to Address (High/Medium/Low) |
|------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------|
| 1    | SQLite database is not suited for high-concurrency production use | Potential write contention and slower performance as the borrower base grows      | High                                  |
| 2    | Single-region deployment with no redundancy                       | Higher latency for users outside the hosting region and a single point of failure | Medium                                |
| 3    | Lender response status is tracked manually by the borrower        | Reduces platform stickiness and delays follow-up on active settlement requests    | Medium                                |

## Scalability Plan:

| S.No | Scalability Aspect | Current State                                                           | Proposed Upgrade / Solution                                                                                        |
|------|--------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Handles ~150 concurrent users in load testing (single backend instance) | Horizontal scaling via containerized FastAPI instances behind a load balancer (e.g., Nginx / AWS ALB)              |
| 2    | Data Storage       | SQLite (single-file, local database)                                    | Migrate to managed PostgreSQL with read replicas to support larger datasets and concurrent access                  |
| 3    | Performance        | Average response ~1.4s; Gemini API calls add the most latency           | Add Redis caching for repeated financial calculations and an async job queue for negotiation letter generation     |
| 4    | Security           | JWT-based authentication over HTTPS                                     | Add rate limiting, refresh token rotation, and periodic security audits/penetration testing as the user base grows |

## Future Roadmap:

| Phase   | Planned Feature / Enhancement                                             | Target Timeline | Expected Impact                                                                      |
|---------|---------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|
| Phase 2 | Automated lender response tracking and multi-language negotiation letters | Q3 2026         | Reduces manual borrower effort and expands reach to non-English speakers             |
| Phase 3 | Mobile app (iOS/Android) and an admin analytics dashboard                 | Q4 2026         | Increases accessibility and gives the team aggregated insight into borrower outcomes |
| Phase 4 | Direct lender API integrations for automated settlement submission        | Q1 2027         | Streamlines the end-to-end settlement process and reduces turnaround time            |